

CERTIPOINT IT SPECIALIST: DATABASES

Complete Study Guide + Sample Questions + Full Mock Exam

Prepared for database exam preparation

How to use this file

Read Parts 1-7 for concepts. Then do the sample questions in Part 8 with explanations.

Take the 45-question mock exam in Part 9 with a 50-minute timer. Do not look at the answer key until finished.

Use Part 11 as a final cheat sheet the night before the exam.

Official exam facts checked

Item	What to expect
Exam	Pearson/Certiport IT Specialist: Databases
Current timed portion	50 minutes
Current question range	35-45 questions for Databases
Question type	Computer-based exam; can include selected-response questions and task-like items. You may need to read SQL, choose the correct SQL, fix SQL, or interpret database design scenarios.
Main skill target	Foundational knowledge of how to design and query relational databases such as MySQL, Microsoft SQL Server, or Oracle.
Important note	This guide uses original practice questions. It does not contain real exam questions.

Sources used: Certiport Exam Lengths page, IT Specialist Certify page, IT Specialist Exam Tutorial PDF, Certiport Practice Tests page, and Credly badge details. Full links are listed in the References section.

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1. How the Certipoint Databases Exam Works

The Databases exam is a timed, proctored, computer-based certification exam. It is not usually a school-style project where you build a full database application from zero. Instead, expect exam questions that test whether you understand database concepts, database design, SQL syntax, SQL results, and common troubleshooting situations.

What happens during the exam

- You sign in through the testing software under a proctor or testing center process.
- A tutorial appears before the real exam timer begins. Read it because it explains navigation, marking questions, and summary review.
- When you select Start Exam, the timer begins.
- You can move back and forward, answer questions, and mark items for review.
- At the exam summary, check unanswered and marked questions before clicking Finish Exam.
- After finishing, your answers are submitted. You cannot change them after submission.

Is it coding, SQL, or multiple choice?

Expect a mixed style. Many questions may look like multiple choice or multiple response, but the content can still be SQL-based. For example, you may be asked to choose the correct query, identify what a query returns, recognize a broken query, or choose the best table design for a scenario.

Study mindset

Do not memorize only definitions. You must be comfortable reading SQL quickly.

You do not need to build a full app, but you should know how to create tables, query data, join tables, filter data, group data, and protect data integrity.

Time strategy for 50 minutes

Time	Action
0-5 min	Answer easy concept questions quickly. Do not overthink definitions.
5-30 min	Handle SQL questions. Read table names, selected columns, filters, and join conditions carefully.

30-42 min	Solve scenario/design questions. Eliminate bad designs and unsafe SQL.
42-50 min	Review marked and unanswered items. Guess intelligently rather than leaving blanks.

2. What to Study First

Priority map

Priority	Topic	Why it matters
High	SELECT, WHERE, ORDER BY, GROUP BY, HAVING	These are the heart of query questions.
High	JOINS and relationships	Database exams often test how tables connect.
High	Primary keys, foreign keys, constraints	Core database design and integrity.
High	DDL and DML	You need to know which SQL commands create objects and which manipulate data.
Medium	Indexes, views, stored procedures	Common database objects.
Medium	Normalization	Helps identify good table design.
Medium	Transactions and ACID	Important for safe updates and deletes.
Medium	Security and permissions	Foundation of database administration.
Low-Medium	Vendor-specific SQL syntax	The exam is foundational, so focus on ANSI-style SQL first.

The fastest way to improve

- Practice explaining every SQL command in plain English.
- Practice predicting query output from small sample tables.
- Memorize the order of SQL clauses: SELECT, FROM, JOIN, WHERE, GROUP BY, HAVING, ORDER BY.
- Learn the difference between WHERE and HAVING.
- Learn when to use INNER JOIN versus LEFT JOIN.
- Learn constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, DEFAULT.

3. Core Database Concepts

Term	Meaning
Database	An organized collection of related data. Example: a school enrollment database.
DBMS	Database Management System. Software used to create, store, manage, and query databases. Examples: MySQL, SQL Server, Oracle, PostgreSQL.
Table	A structure that stores data in rows and columns. Example: Students table.
Row / Record	One complete item in a table. Example: one student.
Column / Field	One attribute of the table. Example: student_id, full_name, course.
Data type	Defines what kind of data a column can hold. Example: INT, VARCHAR, DATE, DECIMAL.
NULL	A missing or unknown value. It is not the same as zero or

	empty string.
Schema	The structure/design of a database: tables, columns, relationships, constraints, and objects.

Keys and relationships

Term	Meaning	Example
Primary Key	Uniquely identifies each row in a table.	student_id in Students
Foreign Key	A column that points to a primary key in another table.	enrollment.student_id references students.student_id
Candidate Key	Any column or set of columns that could uniquely identify a row.	email, student_number
Composite Key	A key made from two or more columns.	student_id + subject_id in Enrollments
Referential Integrity	Ensures relationships stay valid. You cannot reference a non-existing parent row.	Enrollment cannot use student_id 99 if student 99 does not exist.

Relationship types

Relationship	Meaning	Example
One-to-One	One row in table A relates to one row in table B.	Person and passport record
One-to-Many	One row in table A relates to many rows in table B.	One course has many students
Many-to-Many	Many rows in A relate to many rows in B. Usually needs a junction table.	Students and subjects through Enrollments

Normalization

Normalization is the process of organizing data to reduce duplication and improve integrity. For exams, focus mainly on 1NF, 2NF, and 3NF.

Normal Form	Simple meaning	Exam clue
1NF	Each column contains atomic/single values; no repeating groups.	A cell should not contain multiple phone numbers separated by commas.
2NF	Table is in 1NF and every non-key column depends on the whole primary key.	A composite-key table should not store data that depends on only one part of the key.
3NF	Table is in 2NF and non-key columns do not depend on other non-key columns.	Do not store department_name in Students if it depends on department_id.

Bad table design vs better table design

Bad design:

Students(student_id, full_name, subject1, subject2, subject3)

Better design:

Students(student_id, full_name)

Subjects(subject_id, subject_name)

Enrollments(student_id, subject_id)

4. SQL Command Guide

SQL command categories

Category	Purpose	Common commands
DDL	Defines or changes database objects.	CREATE, ALTER, DROP, TRUNCATE
DML	Manipulates table data.	INSERT, UPDATE, DELETE
DQL	Retrieves data.	SELECT
DCL	Controls permissions.	GRANT, REVOKE
TCL	Controls transactions.	COMMIT, ROLLBACK, SAVEPOINT

Sample practice schema

Use this schema for the practice questions and mock exam:

```
CREATE TABLE departments (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(80) NOT NULL
);
```

```
CREATE TABLE students (
    student_id INT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    department_id INT,
    year_level INT,
    email VARCHAR(120) UNIQUE,
    FOREIGN KEY (department_id) REFERENCES departments(department_id)
);
```

```
CREATE TABLE courses (
    course_id INT PRIMARY KEY,
    course_code VARCHAR(20) UNIQUE NOT NULL,
    course_title VARCHAR(100) NOT NULL,
    units INT CHECK (units > 0)
);
```

```
CREATE TABLE enrollments (
    enrollment_id INT PRIMARY KEY,
    student_id INT NOT NULL,
    course_id INT NOT NULL,
    grade DECIMAL(4,2),
    enrollment_date DATE,
    FOREIGN KEY (student_id) REFERENCES students(student_id),
    FOREIGN KEY (course_id) REFERENCES courses(course_id)
);
```

SELECT basics

```
SELECT full_name, year_level
FROM students
WHERE department_id = 1
ORDER BY full_name ASC;
```

Clause	What it does
SELECT	Chooses the columns to display.
FROM	Chooses the source table.

WHERE	Filters individual rows before grouping.
ORDER BY	Sorts the result. ASC is ascending; DESC is descending.

WHERE operators

Operator	Use
=	Equal to
<> or !=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
BETWEEN	Within a range
IN	Matches any value in a list
LIKE	Pattern matching
IS NULL	Tests for missing value
AND / OR	Combines conditions

Aggregation and grouping

```
SELECT department_id, COUNT(*) AS total_students
FROM students
GROUP BY department_id
HAVING COUNT(*) >= 5;
```

Remember: WHERE filters rows before grouping. HAVING filters groups after aggregation.

JOINS

Join type	Meaning	Exam clue
INNER JOIN	Returns rows that have matching values in both tables.	Use when you only want students with valid related rows.
LEFT JOIN	Returns all rows from the left table plus matching rows from the right table.	Use when you want all students even if they have no enrollments.
RIGHT JOIN	Returns all rows from the right table plus matching rows from the left table.	Less common; can usually rewrite as LEFT JOIN.
CROSS JOIN	Returns every combination of rows.	Usually huge result; watch for accidental missing join condition.

```
SELECT s.full_name, c.course_code, e.grade
FROM students s
INNER JOIN enrollments e ON s.student_id = e.student_id
INNER JOIN courses c ON e.course_id = c.course_id;
```

INSERT, UPDATE, DELETE

```
INSERT INTO departments (department_id, department_name)
VALUES (1, 'Department of Computing Education');
```

```
UPDATE students
SET year_level = 3
WHERE student_id = 1001;
```

```
DELETE FROM enrollments
WHERE enrollment_id = 10;
```

Dangerous exam trap

UPDATE without WHERE can update every row in the table.
DELETE without WHERE can delete every row in the table.

Creating tables with constraints

```
CREATE TABLE books (
  book_id INT PRIMARY KEY,
  title VARCHAR(150) NOT NULL,
  isbn VARCHAR(20) UNIQUE,
  price DECIMAL(10,2) CHECK (price >= 0),
  status VARCHAR(20) DEFAULT 'Available'
);
```

5. Database Objects and Design

Object	Purpose	Simple example
Table	Stores rows and columns.	students
View	Saved SELECT query that behaves like a virtual table.	active_students_view
Index	Improves search speed, especially on frequently filtered columns.	index on email
Stored Procedure	Saved SQL routine that can run multiple statements.	AddStudent()
Trigger	Automatically runs when INSERT, UPDATE, or DELETE occurs.	after insert audit log
Constraint	Rule that protects data correctness.	NOT NULL, FOREIGN KEY

Views

```
CREATE VIEW student_course_view AS
SELECT s.full_name, c.course_code, e.grade
FROM students s
JOIN enrollments e ON s.student_id = e.student_id
JOIN courses c ON e.course_id = c.course_id;
```

A view is useful when you repeatedly need the same query or want to hide sensitive columns.

Stored procedures

```
-- MySQL-style example
DELIMITER //
CREATE PROCEDURE GetStudentsByDepartment(IN p_department_id INT)
BEGIN
  SELECT full_name, year_level
  FROM students
  WHERE department_id = p_department_id;
END //
DELIMITER ;

CALL GetStudentsByDepartment(1);
```

Stored procedures are reusable SQL routines stored in the database. They can accept parameters, perform multi-step operations, and centralize database logic.

Indexes

```
CREATE INDEX idx_students_email
ON students(email);
```

Indexes make lookups faster but can slow down INSERT, UPDATE, and DELETE because the index also needs to be maintained.

6. Data Integrity, Security, and Administration

Data integrity

Integrity type	Meaning	How to enforce
Entity integrity	Every row must be uniquely identifiable.	PRIMARY KEY
Referential integrity	Foreign keys must point to valid parent rows.	FOREIGN KEY
Domain integrity	Column values must follow allowed type/range/rules.	Data type, CHECK, NOT NULL
User-defined integrity	Business-specific rules.	CHECK constraints, triggers, procedures, application validation

Transactions and ACID

```
START TRANSACTION;
UPDATE accounts SET balance = balance - 500 WHERE account_id = 1;
UPDATE accounts SET balance = balance + 500 WHERE account_id = 2;
COMMIT;
```

ACID property	Meaning
Atomicity	All steps succeed or all are undone.
Consistency	Database moves from one valid state to another valid state.
Isolation	Concurrent transactions do not incorrectly interfere with each other.
Durability	Committed changes remain saved even after failure.

Security basics

- Use least privilege: give users only the permissions they need.
- Use roles/groups instead of assigning many individual permissions manually.
- Protect sensitive data with access control and, when appropriate, encryption.
- Use parameterized queries in applications to reduce SQL injection risk.
- Back up databases regularly and test restore procedures.
- Audit important operations such as login failures, permission changes, and sensitive data access.

7. Troubleshooting SQL and Database Problems

Problem	Possible cause	How to fix
Syntax error near FROM	Missing comma or wrong clause order.	Check SELECT list and SQL clause order.

Unknown column	Column name misspelled or table alias missing.	Check schema and aliases.
Foreign key constraint fails	Child row references parent row that does not exist.	Insert parent row first or correct the foreign key value.
Duplicate key error	PRIMARY KEY or UNIQUE value already exists.	Use a different key or update existing row.
Query returns duplicate rows	Join condition is incomplete or many-to-many relationship.	Check ON condition and whether DISTINCT/GROUP BY is appropriate.
Aggregate error	Selected non-aggregated column is not in GROUP BY.	Group by non-aggregated columns or aggregate them.
Wrong result with NULL	Using = NULL instead of IS NULL.	Use IS NULL or IS NOT NULL.
Accidental all-row update/delete	Missing WHERE clause.	Use transactions, backups, and WHERE filters.

8. Sample Questions with Explanations

These are warm-up questions with answers immediately shown. The full mock exam comes after this section.

1. Which SQL clause filters rows before grouping?

- A. HAVING
- B. WHERE
- C. ORDER BY
- D. GROUP BY

Answer: B

Why: WHERE filters individual rows before GROUP BY. HAVING filters groups after aggregation.

2. Which constraint prevents two students from having the same email?

- A. NOT NULL
- B. FOREIGN KEY
- C. UNIQUE
- D. CHECK

Answer: C

Why: UNIQUE ensures values in a column are not duplicated.

3. Which command is DDL?

- A. INSERT
- B. UPDATE
- C. CREATE TABLE
- D. SELECT

Answer: C

Why: DDL defines database objects. CREATE TABLE creates a table.

4. What does COUNT(*) return?

- A. The number of columns
- B. The number of rows
- C. The largest value
- D. The number of primary keys

Answer: B

Why: COUNT(*) counts rows in the result group or result set.

5. Which join returns all rows from the left table even without matches in the right table?

- A. INNER JOIN
- B. LEFT JOIN
- C. CROSS JOIN
- D. SELF JOIN

Answer: B

Why: LEFT JOIN preserves all rows from the left table.

6. Which normal form removes repeating groups and requires atomic values?

- A. 1NF
- B. 2NF
- C. 3NF
- D. BCNF

Answer: A

Why: 1NF requires each cell to hold a single atomic value.

7. A foreign key mainly enforces which integrity?

- A. Entity integrity
- B. Referential integrity
- C. Domain integrity
- D. File integrity

Answer: B

Why: Foreign keys make sure child rows reference valid parent rows.

8. Which SQL is safest for changing only student 1001?

- A. UPDATE students SET year_level = 3;
- B. UPDATE students SET year_level = 3 WHERE student_id = 1001;
- C. ALTER students SET year_level = 3;
- D. MODIFY students year_level = 3;

Answer: B

Why: The WHERE clause limits the update to one intended student.

9. Which command saves a transaction permanently?

- A. ROLLBACK
- B. COMMIT
- C. SAVEPOINT
- D. UNDO

Answer: B

Why: COMMIT makes the transaction changes durable.

10. What does SELECT DISTINCT department_id FROM students do?

- A. Deletes duplicate departments
- B. Returns only unique department_id values
- C. Sorts departments
- D. Counts departments

Answer: B

Why: DISTINCT removes duplicate rows from the query result.

9. Full 45-Question Mock Exam

Mock exam rules

Set a 50-minute timer.

Answer all 45 questions. Some questions are multiple answer and will say "Choose TWO".

Use scratch paper for query output questions.

Do not check the answer key until finished.

1. Which statement best describes a relational database?

- A. A set of unrelated files stored in folders
- B. A database that organizes data into related tables
- C. A spreadsheet that cannot enforce rules
- D. A programming language used only for reports

2. Which object stores data in rows and columns?

- A. View
- B. Index
- C. Table
- D. Trigger

3. Which column is the best primary key for a Students table?

- A. full_name
- B. student_id
- C. course_name
- D. year_level

4. A table Orders has customer_id that references Customers(customer_id). What is Orders.customer_id?

- A. Primary key
- B. Foreign key
- C. Composite database
- D. Index only

5. Which SQL command retrieves data?

- A. SELECT
- B. INSERT
- C. DROP
- D. ALTER

6. Which SQL clause sorts query results?

- A. WHERE
- B. ORDER BY
- C. GROUP BY
- D. HAVING

7. Which condition correctly tests for missing email values?

- A. email = NULL
- B. email IS NULL
- C. email == NULL
- D. email LIKE NULL

8. Choose TWO commands that are DML.

- A. INSERT
- B. CREATE
- C. UPDATE
- D. DROP

9. Which constraint requires a column to always have a value?

- A. UNIQUE
- B. NOT NULL
- C. FOREIGN KEY
- D. DEFAULT

10. Which SQL statement adds a new row?

- A. ADD ROW students VALUES (...)
- B. INSERT INTO students (...) VALUES (...)
- C. CREATE ROW students (...)
- D. ALTER TABLE students VALUES (...)

11. What is the main purpose of normalization?

- A. Make every table have only one row
- B. Reduce redundancy and improve data integrity
- C. Encrypt all database files
- D. Make queries impossible to change

12. Which design best represents a many-to-many relationship between Students and Courses?

- A. Put course1, course2, course3 columns in Students
- B. Use Students, Courses, and Enrollments junction table
- C. Store all student and course information in one table
- D. Use only the Courses table

13. Which SQL creates a table?

- A. CREATE TABLE
- B. MAKE TABLE
- C. INSERT TABLE
- D. SELECT TABLE

14. Which query returns students in year level 3?

- A. SELECT * FROM students WHERE year_level = 3;
- B. SELECT * FROM students HAVING year_level = 3;
- C. GET students WHERE year_level = 3;
- D. SELECT year_level = 3 FROM students;

15. Which aggregate function returns the average grade?

- A. COUNT
- B. SUM
- C. AVG
- D. MAX

16. In SQL, which clause filters groups after GROUP BY?

- A. WHERE
- B. HAVING
- C. ORDER
- D. FILTER BY

17. What does this query do? SELECT department_id, COUNT(*) FROM students GROUP BY department_id;

- A. Counts students per department
- B. Deletes duplicate departments
- C. Shows only departments with one student
- D. Creates a department table

18. Which join returns only matching rows from both tables?

- A. LEFT JOIN
- B. RIGHT JOIN
- C. INNER JOIN
- D. FULL OUTER JOIN

19. What is likely wrong with this query? SELECT full_name FROM students ORDER full_name;

- A. SELECT is misspelled
- B. ORDER BY should be used instead of ORDER
- C. FROM is not allowed
- D. full_name cannot be selected

20. Which query returns unique department IDs from students?

- A. SELECT UNIQUE department_id FROM students;
- B. SELECT DISTINCT department_id FROM students;
- C. SELECT ONLY department_id FROM students;
- D. SELECT DIFFERENT department_id FROM students;

21. Choose TWO true statements about indexes.

- A. They can improve search performance
- B. They always remove the need for primary keys
- C. They can add overhead to inserts and updates
- D. They store unrelated application code

22. What is a view?

- A. A saved query that can be used like a virtual table
- B. A database backup file
- C. A table with no columns
- D. A command that deletes records

23. Which statement is dangerous because it changes all rows?

- A. UPDATE students SET year_level = 3 WHERE student_id = 7;
- B. UPDATE students SET year_level = 3;
- C. SELECT * FROM students;
- D. INSERT INTO students(student_id) VALUES(7);

24. Which action undoes uncommitted transaction changes?

- A. COMMIT
- B. ROLLBACK
- C. GRANT
- D. CREATE

25. Which ACID property means a transaction is all-or-nothing?

- A. Atomicity
- B. Consistency
- C. Isolation
- D. Durability

26. Which command gives a user permission?

- A. REVOKE
- B. GRANT
- C. DENY TABLE
- D. ALLOW ALL

27. What is the principle of least privilege?

- A. Give every user admin access
- B. Give users only the access needed for their work
- C. Remove all passwords
- D. Store all permissions in one text file

28. Which data type is most appropriate for a birth date?

- A. VARCHAR(255)
- B. DATE
- C. DECIMAL(10,2)
- D. BOOLEAN

29. Which data type is most appropriate for money values?

- A. DECIMAL(10,2)
- B. TEXT
- C. DATE
- D. CHAR(1)

30. Which constraint checks that units are greater than zero?

- A. CHECK (units > 0)
- B. DEFAULT (units > 0)
- C. PRIMARY KEY (units > 0)
- D. NOT NULL (units > 0)

31. What causes a foreign key constraint error?

- A. Trying to reference a parent row that does not exist
- B. Selecting too many columns
- C. Using ORDER BY
- D. Using COUNT(*)

32. Which query lists student names and course codes using the schema in this guide?

- A. SELECT full_name, course_code FROM students, courses;
- B. SELECT s.full_name, c.course_code FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id;
- C. SELECT students JOIN courses;
- D. GET full_name, course_code FROM students;

33. What is a stored procedure?

- A. A saved SQL routine stored in the database
- B. A file attachment inside a row
- C. A backup of a table only
- D. A command that only sorts data

34. What is a trigger?

- A. A routine that automatically runs in response to database events
- B. A manual SELECT query only
- C. A table with no primary key
- D. A command to shut down the database

35. Which SQL clause limits rows before GROUP BY?

- A. WHERE
- B. HAVING
- C. ORDER BY
- D. GROUP BY

36. Which query returns courses with average grade greater than 90?

- A. SELECT course_id FROM enrollments WHERE AVG(grade) > 90 GROUP BY course_id;
- B. SELECT course_id FROM enrollments GROUP BY course_id HAVING AVG(grade) > 90;
- C. SELECT course_id HAVING grade > 90 FROM enrollments;
- D. SELECT AVG(course_id) FROM enrollments WHERE grade > 90;

37. Which result does a CROSS JOIN produce?

- A. Only matching rows
- B. All combinations of rows from both tables
- C. Only rows with NULL values
- D. Only duplicate rows

38. Choose TWO examples of constraints.

- A. PRIMARY KEY
- B. FOREIGN KEY
- C. SELECT
- D. ORDER BY

39. Which normal form deals with removing transitive dependencies?

- A. 1NF
- B. 2NF
- C. 3NF
- D. 0NF

40. A column department_name depends on department_id, not directly on student_id. If both are stored in Students, what issue may exist?

- A. Transitive dependency
- B. Perfect normalization
- C. No possible redundancy
- D. A cross join

41. Which command removes a table structure and its data?

- A. DELETE FROM table_name
- B. DROP TABLE table_name
- C. REMOVE ROW table_name
- D. ROLLBACK TABLE table_name

42. Which command removes rows but keeps the table structure?

- A. DELETE
- B. DROP TABLE
- C. CREATE
- D. GRANT

43. Which is the best way to prevent SQL injection in application code?

- A. Use parameterized queries
- B. Make all fields VARCHAR
- C. Remove all indexes
- D. Use SELECT * always

44. Which backup-related statement is best?

- A. Backups are useful only if restore has been tested
- B. A backup is unnecessary when using primary keys
- C. Backups replace permissions
- D. A view is the same as a backup

45. You need all students even if they are not enrolled in any course. Which join should start from Students?

- A. INNER JOIN
- B. LEFT JOIN
- C. CROSS JOIN
- D. SELF JOIN

10. Answer Key with Explanations

#	Answer	Explanation
1	B	Relational databases organize related data in tables.
2	C	Tables store data in rows and columns.
3	B	student_id is stable and unique; names may repeat.
4	B	A foreign key references a primary key in another table.
5	A	SELECT retrieves data.
6	B	ORDER BY sorts results.
7	B	NULL must be checked using IS NULL.
8	A and C	INSERT and UPDATE are DML commands.
9	B	NOT NULL requires a value.
10	B	INSERT INTO ... VALUES ... adds a new row.
11	B	Normalization reduces duplication and improves integrity.
12	B	A junction table correctly represents many-to-many relationships.
13	A	CREATE TABLE creates a new table.
14	A	WHERE filters rows by year_level.
15	C	AVG returns average.
16	B	HAVING filters groups after GROUP BY.
17	A	The query groups by department and counts rows per group.
18	C	INNER JOIN returns only matching rows.
19	B	The correct clause is ORDER BY.
20	B	DISTINCT returns unique result rows.
21	A and C	Indexes can speed lookups but add write overhead.
22	A	A view is a saved SELECT query/virtual table.
23	B	No WHERE means every row is updated.
24	B	ROLLBACK undoes uncommitted changes.
25	A	Atomicity means all-or-nothing.
26	B	GRANT gives permissions.
27	B	Least privilege means minimal necessary access.
28	B	DATE is best for dates.
29	A	DECIMAL is best for exact money values.

30	A	CHECK can enforce a value rule.
31	A	Foreign key requires valid referenced parent value.
32	B	Correct join path is students -> enrollments -> courses.
33	A	Stored procedures are saved routines.
34	A	Triggers run automatically after/before database events.
35	A	WHERE filters rows before grouping.
36	B	Aggregates in filters belong in HAVING.
37	B	CROSS JOIN returns Cartesian product.
38	A and B	Primary and foreign keys are constraints.
39	C	3NF removes transitive dependencies.
40	A	department_name depends on department_id, creating transitive dependency.
41	B	DROP TABLE removes table definition and data.
42	A	DELETE removes rows but keeps the table.
43	A	Parameterized queries separate code from data.
44	A	A backup is valuable when it can actually be restored.
45	B	LEFT JOIN keeps all rows from Students.

11. Final SQL Cheat Sheet

SQL clause order

```

SELECT columns
FROM table
JOIN other_table ON condition
WHERE row_condition
GROUP BY grouped_columns
HAVING group_condition
ORDER BY sort_columns;
```

Must-know patterns

```

-- Select with filter
SELECT * FROM students WHERE year_level = 3;

-- Sort
SELECT * FROM students ORDER BY full_name ASC;

-- Count per group
SELECT department_id, COUNT(*) AS total
FROM students
GROUP BY department_id;

-- Filter groups
SELECT course_id, AVG(grade) AS avg_grade
```

```
FROM enrollments
GROUP BY course_id
HAVING AVG(grade) >= 90;
```

```
-- Inner join
SELECT s.full_name, c.course_code
FROM students s
JOIN enrollments e ON s.student_id = e.student_id
JOIN courses c ON e.course_id = c.course_id;
```

```
-- Left join
SELECT s.full_name, e.course_id
FROM students s
LEFT JOIN enrollments e ON s.student_id = e.student_id;
```

```
-- Insert
INSERT INTO students(student_id, full_name, department_id, year_level)
VALUES (1001, 'Aljun Cursiga', 1, 3);
```

```
-- Update safely
UPDATE students
SET year_level = 4
WHERE student_id = 1001;
```

```
-- Delete safely
DELETE FROM students
WHERE student_id = 1001;
```

Quick definitions to memorize

Term	One-line definition
Primary key	Uniquely identifies each record.
Foreign key	References a primary key in another table.
Index	Speeds up searching but adds write overhead.
View	Saved query/virtual table.
Stored procedure	Saved SQL routine.
Trigger	Automatic routine fired by events.
Transaction	Group of SQL operations treated as one unit.
Commit	Save transaction changes.
Rollback	Undo uncommitted transaction changes.
Normalization	Organize data to reduce redundancy and improve integrity.

12. 7-Day and 14-Day Study Plan

7-day crash plan

Day	Focus	Tasks
1	Database basics	Tables, rows, columns, data types, keys, relationships. Make flashcards.
2	SQL SELECT	Practice WHERE, ORDER BY, LIKE, IN, BETWEEN, NULL checks.
3	JOINS	Practice INNER JOIN and LEFT JOIN using the sample schema.
4	Aggregates	Practice COUNT, SUM, AVG, MIN, MAX,

		GROUP BY, HAVING.
5	DDL/DML and constraints	CREATE TABLE, INSERT, UPDATE, DELETE, PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK.
6	Administration/security	Transactions, ACID, permissions, backups, least privilege, SQL injection.
7	Mock exam	Take the 45-question mock exam with a timer. Review weak areas.

14-day comfortable plan

Day(s)	Focus
1-2	Core database terms, data types, primary keys, foreign keys, relationships.
3-4	SELECT, WHERE, ORDER BY, filtering, NULL, LIKE, IN, BETWEEN.
5-6	JOINS and relationship-based query questions.
7-8	GROUP BY, HAVING, aggregate functions, interpreting query output.
9	CREATE TABLE, constraints, views, indexes, stored procedures, triggers.
10	INSERT, UPDATE, DELETE, transactions, COMMIT, ROLLBACK.
11	Normalization and table design.
12	Security, roles, permissions, backups, SQL injection.
13	Troubleshooting syntax errors, runtime errors, and design mistakes.
14	Take mock exam. Review all wrong answers and retake missed questions.

Final night checklist

- I can explain primary key vs foreign key clearly.
- I can write a basic SELECT query with WHERE and ORDER BY.
- I know WHERE vs HAVING.
- I know INNER JOIN vs LEFT JOIN.
- I know CREATE TABLE with constraints.
- I know INSERT, UPDATE, DELETE and the danger of missing WHERE.
- I know what COMMIT and ROLLBACK do.
- I know the purpose of views, indexes, stored procedures, and triggers.
- I know the purpose of normalization and 1NF/2NF/3NF.
- I can complete 45 questions in 50 minutes.

References

Certiport Exam Lengths page: <https://certiport.pearsonvue.com/Educator-resources/Exam-details/Exam-lengths.aspx>

Certiport IT Specialist Certify page: <https://certiport.pearsonvue.com/Certifications/ITSpecialist/Certification/Certify.aspx>

Certiport IT Specialist Exam Tutorial PDF: [https://certiport.pearsonvue.com/Educator-resources/Exam-details/Exam-tutorials/IT Specialist Tutorial.pdf](https://certiport.pearsonvue.com/Educator-resources/Exam-details/Exam-tutorials/IT%20Specialist%20Tutorial.pdf)

Certiport Practice Tests page: <https://certiport.pearsonvue.com/Certifications/ITSpecialist/Certification/Practice.aspx>

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